

SIBIU, Romania

WMO: 152600

Lat: 45.7894N

Lon: 24.0914E

Elev: 444

StdP: 96.10

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-12.8	-18.1	0.8	-15.3	-14.8	1.1	-11.8	10.3	5.3	8.9	5.5	0.9	220	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	31.3	19.7	29.6	19.5	28.0	18.8	21.5	28.0	20.5	26.9	19.7	25.9	3.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.9	23.9	18.4	14.0	23.0	17.7	13.3	22.3	64.3	28.1	61.1	27.0	58.3	26.0	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.7	6.3	DB	-21.3	33.9	4.0	2.2	-24.2	35.4	-26.5	36.7	-28.8	37.9	-31.7	39.5	(4)
(5)			WB	-21.5	23.1	4.0	1.1	-24.3	23.8	-26.7	24.5	-28.9	25.1	-31.8	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-2.2	0.5	4.8	10.5	15.2	18.7	20.3	20.2	15.4	10.0	4.9	-0.8	(6)	
	DBStd	8.95	5.19	5.31	4.56	3.93	3.31	3.16	2.82	2.87	3.62	4.10	4.79	5.39	(7)	
	HDD10.0	1407	377	267	166	40	3	0	0	0	2	51	166	335	(8)	
	HDD18.3	3311	636	499	418	236	105	34	13	14	100	258	404	593	(9)	
	CDD10.0	1351	0	1	6	54	164	261	321	316	165	52	11	0	(10)	
	CDD18.3	213	0	0	0	1	8	44	76	72	13	1	0	0	(11)	
	CDH23.3	2229	0	0	1	17	119	453	740	722	164	13	0	0	(12)	
	CDH26.7	659	0	0	0	1	13	118	240	247	37	2	0	0	(13)	
(14) Wind	WSAvg	2.1	1.7	2.0	2.5	2.6	2.3	2.2	2.2	2.0	2.0	1.9	2.0	1.7	(14)	
(15) Precipitation	PrecAvg	654	27	25	33	54	79	98	93	71	53	43	34	31	(15)	
	PrecMax	875	72	53	87	129	171	233	226	227	186	147	91	82	(16)	
	PrecMin	361	5	0	3	5	31	3	13	3	1	2	1	1	(17)	
	PrecStd	112	16	15	20	27	37	48	44	49	38	29	19	19	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	15.8	20.2	25.5	28.2	31.7	33.9	33.8	30.3	24.8	20.2	13.9	(19)	
		MCWB	7.8	9.3	10.6	14.8	17.0	19.9	19.6	19.6	17.7	15.7	12.6	8.8	(20)	
	2%	DB	9.3	12.5	17.4	22.3	26.1	29.8	31.1	31.5	27.3	22.5	17.5	11.0	(21)	
		MCWB	6.4	7.6	9.6	13.3	16.6	19.8	20.0	19.8	17.2	14.6	11.5	7.4	(22)	
	5%	DB	6.8	10.2	15.0	20.0	24.5	27.9	29.3	29.4	25.0	20.0	15.2	8.4	(23)	
		MCWB	4.6	6.3	8.6	12.1	16.2	19.0	19.8	19.4	16.3	13.9	10.5	5.9	(24)	
	10%	DB	4.6	8.0	12.7	18.1	22.7	26.1	27.4	27.4	22.9	17.6	12.8	6.2	(25)	
		MCWB	3.0	5.1	7.5	11.1	15.3	18.3	19.0	18.6	15.8	12.6	9.2	4.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	9.6	11.8	15.7	19.2	22.3	22.5	22.7	19.6	17.3	13.5	9.4	(27)	
		MCDB	11.3	14.7	18.5	23.6	25.6	28.7	29.0	29.5	26.7	23.2	18.4	12.6	(28)	
	2%	WB	6.5	8.0	10.2	14.2	17.8	20.8	21.4	21.3	18.0	15.6	12.1	7.7	(29)	
		MCDB	9.1	12.0	15.9	20.9	23.9	27.2	27.7	28.2	24.6	20.7	16.6	10.6	(30)	
	5%	WB	4.7	6.7	9.1	12.9	16.7	19.8	20.6	20.2	17.1	14.3	10.8	6.1	(31)	
		MCDB	6.5	9.7	14.4	18.7	22.8	26.1	26.9	26.5	23.3	18.9	14.8	8.3	(32)	
	10%	WB	3.0	5.3	7.8	11.8	15.8	18.9	19.8	19.4	16.3	13.1	9.5	4.7	(33)	
		MCDB	4.3	7.8	12.1	16.6	21.6	24.9	25.8	25.5	21.5	17.1	12.6	6.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	9.2	10.5	11.4	11.6	11.7	12.0	12.4	11.6	11.8	9.9	7.4	(35)	
		MCDBR	10.1	12.6	15.1	14.9	14.9	14.8	15.1	15.7	15.4	15.9	13.9	10.6	(36)	
	5% WB	MCWBR	7.3	8.0	8.4	7.5	7.2	6.6	6.3	6.1	6.7	8.8	8.5	7.3	(37)	
		MCWBR	9.6	11.7	13.7	13.5	13.6	13.3	13.7	13.6	13.6	14.1	12.8	9.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.308	0.337	0.379	0.439	0.424	0.420	0.439	0.450	0.403	0.364	0.337	0.308	(40)	
		taud	2.345	2.276	2.225	2.096	2.210	2.257	2.212	2.174	2.275	2.380	2.399	2.376	(41)	
	Ebn at Noon		788	829	841	817	846	849	827	799	807	791	747	744	(42)	
		Edh at Noon	76	99	121	150	139	133	137	138	114	89	73	67	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.33	2.15	3.24	4.31	5.21	5.72	5.71	5.21	3.76	2.61	1.52	1.08	(44)	
	RadStd		0.16	0.27	0.39	0.47	0.50	0.52	0.49	0.47	0.39	0.32	0.18	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.79	N/A	N/A	N/A	N/A	N/A	-151	-221	+120	+50
Regional (75 neighbors)		+0.67	N/A	N/A	+0.89	N/A	N/A	-134	-195	+85	+19

Nomenclature: See separate page